Venkata Lakshmi Parimala Pasupuleti

AI Tech Specialist · Python · Generative AI · Automation · Data Analytics · Workflow Optimization · Responsible AI · 9yr Production

+1-703-261-5410 | lakshmiparimalap@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | github.com/venkatalakshmiparimala | Virginia, USA

PROFESSIONAL SUMMARY

AI and technology specialist with 9+ years of experience developing and implementing AI solutions, automation systems, and data-driven tools that improve organizational efficiency and service delivery. I build practical AI applications that solve real problems - at Magizhs, I designed and deployed a production AI system for an external organization end-to-end, translating ambiguous requirements into working automation delivered in weeks. At George Mason University, I supported 600+ students and faculty in using Python, SQL, and data analysis tools effectively - identifying where technology helps and where human judgment matters more. At Entrata, I built automation pipelines that eliminated 80% of manual operational workload and delivered data insights that directly improved financial process accuracy across 100K+ housing units. I work with generative AI tools daily (Python, LangChain, OpenAI API, Claude Code) and apply responsible AI practices - human-in-the-loop controls, audit logging, privacy-by-design, and bias awareness - as first-class requirements in every system I build. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

AI & Automation: Python, Generative AI, LangChain, LangGraph, OpenAI API, Anthropic API, Agentic AI, Workflow Automation, Process Improvement, RAG Pipelines, NLP, BERT, spaCy, Prompt Engineering, Responsible AI, AI Ethics, Human-in-the-Loop, Privacy & Security

Data & Analytics: SQL, PostgreSQL, MySQL, Oracle, Data Analysis, Statistical Modeling, PyTorch, scikit-learn, Pandas, NumPy, Tableau, Data Visualization, ETL Pipelines, R, Excel/Power BI

Tools & Delivery: FastAPI, REST APIs, Docker, AWS (S3, EC2, RDS), GitHub Actions CI/CD, Automated Testing, TypeScript, React.js, Linux, Bash, Git, Agile/Scrum, Staff Training & Support, Technical Communication

PROJECTS

AgentProbe: AI Evaluation & Automation Platform

Generative AI · Automation · Responsible AI · Open Source

Python · FastAPI · LangGraph · GitHub Actions · OpenAI API · Anthropic API | github.com/venkatalakshmiparimala/AgentProbe

Open-source platform for evaluating and automating AI workflows - built to help organizations adopt generative AI responsibly and reliably. Multi-provider AI integration (OpenAI, Anthropic, Ollama) via clean abstractions; 5-metric quality engine tracking correctness, accuracy, and hallucination rate to ensure AI outputs meet standards before deployment; automated testing that catches quality regressions. Built with privacy, security, and ethical AI practices as first-class requirements throughout.

Leasel: Responsible AI System - Human Review & Audit Logging

Responsible AI · HITL · Privacy · Compliance

Python · LangGraph · FastAPI · OCR · SQLite · Claude Code | github.com/venkatalakshmiparimala/Lease_complaince_rag

AI system built with responsible AI practices at the core - automated document processing and AI-powered analysis with mandatory human review escalation for high-stakes decisions; deterministic guardrails ensure AI never makes regulated decisions autonomously; full audit logging for every decision path; privacy-by-design data handling throughout. Demonstrates how to implement AI tools that improve efficiency while keeping humans in control of decisions that matter. 93.6% precision across 1,240 test cases.

AI on FHIR: Workflow Automation - Reducing Manual Retrieval by 60%

Workflow Automation · Data Insights · Privacy

Python · Flask · React.js · TypeScript · Docker · BERT · spaCy · LLMs · REST APIs | github.com/venkatalakshmiparimala/AI_on_FHIR

AI-powered workflow automation that reduced manual data retrieval time by ~60% - translates natural-language questions into structured data queries using NLP and LLMs, making information instantly accessible without technical expertise. HIPAA-compliant with role-based access control and full data isolation. Demonstrates the practical benefit of AI tools for non-technical staff: faster access to insights, less manual work, privacy and security maintained throughout.

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer - AI Implementation & Automation

- Designed and deployed a production AI automation system for an external organization - gathered requirements from stakeholders, identified the highest-value workflow to automate, built and tested the solution, and deployed to production in weeks as sole technical owner. Trained staff on using the system effectively and documented workflows for ongoing operation.

George Mason University

Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Supported 600+ students in learning Python, SQL, and data analysis tools - identifying where individuals struggled, adapting explanations for different learning styles, and providing one-on-one technical coaching. Built ML pipelines (R^2=0.977) and designed data visualization dashboards that improved research comprehension 40% for non-technical faculty stakeholders.

Entrata - Apartment & Affordable Housing SaaS

Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **Workflow automation and efficiency** - built Python and Bash automation pipelines that eliminated 80% of manual operational workload; AWS data pipelines processing 180,000 records for housing eligibility and payment calculations; 200+ SQL optimizations (30% performance gain); internal tools reducing triage time 40%.
- **Data insights and reporting** - owned financial data pipelines for compliance reporting across 100K+ housing units; delivered data-driven insights to product, operations, and compliance stakeholders across 120+ Agile sprints; 20+ full-stack production modules shipped end-to-end.
- **Staff support and responsible technology use** - mentored 5+ engineers on best practices; 300+ peer code reviews; 100% PHI/HIPAA compliance with privacy-by-design; zero security incidents across 7yr; maintained awareness of data privacy, security, and ethical use of technology throughout.

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built data workflows (PHP, Python, PostgreSQL) for housing management; AWS S3 migration; automation pipelines for 1.6M records.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017